PLOT: Phase-Level Objective Tilting with Learned Adversarial Motion Rewards

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***Abstract*— Learning feedback controllers from reference motion often relies on carefully weighted pose, velocity, root, end-effector, and contact objectives. Learned adversarial rewards remove this manual feature-wise scalarization, yet standard expected-return optimization still allows frequently rewarded phases to compensate for brief but decisive tracking failures. We present Phase-Level Objective Tilting (PLOT), a phase-aware batch surrogate for learned-reward motion control. PLOT evaluates reference phases with the same base adversarial reward and exponentially tilts the empirical phase distribution toward low-utility regions under relative-entropy regularization. The resulting density ratio reweights rewards before advantage estimation, without introducing a manually designed phase reward, semantic phase supervision, a critic beyond the standard PPO value function, or a phase index in the policy. Across motions involving locomotion, rotation, ground contact, recovery, striking, and object interaction, [TBD: insert the final suite-level completion and tracking result].**

I. Introduction

Learning feedback controllers that reproduce reference motions under contact, inertia, and actuation constraints remains a central problem in character and humanoid control. Most existing methods define tracking quality by combining pose, velocity, root, end-effector, and contact errors with manually selected scales and weights [1]. These choices must reconcile quantities with different units and often require retuning across motions.

Adversarial imitation offers a way to learn the tracking objective instead of manually specifying its feature weights. A discriminator can aggregate synchronized tracking discrepancies and provide a single reward to the policy. This removes a substantial source of motion-dependent reward engineering. Yet it does not determine how rewards from different times should be combined. Standard reinforcement learning maximizes an expected discounted sum. Consequently, high reward over frequent or easy phases can compensate for low reward over a brief event that is indispensable to the motion. A controller may therefore achieve a competitive average while missing the transition that defines the skill.

This distinction exposes two different scalarization problems. The first is *feature-wise* aggregation of errors in heterogeneous physical quantities at one instant; a learned adversarial reward can address it. The second is *phase-wise* aggregation, which determines how tracking quality at different positions in the reference motion influences the policy update. An ordinary temporal average silently makes this choice through rollout occupancy. It favors phases that occur often and permits compensation across time, even when the reference is only complete if every decisive phase is executed.

We introduce PLOT to address temporal compensation without redesigning the learned reward. From a frozen rollout, PLOT forms an entropy-regularized soft minimum of phase utilities. Its closed-form tilt emphasizes lower-utility phases while remaining anchored to observed occupancy. We then derive an occupancy-corrected batch reward whose immediate phase average matches this adversarial aggregation; proximal policy optimization (PPO) applies its usual discounted-return surrogate to that reward. The policy and discriminator receive no additional phase index, and the method requires neither a critic beyond the standard PPO value function nor a manually designed phase reward.

Our contributions are:

- We separate feature-wise reward aggregation from phase-wise temporal compensation in learned-reward motion tracking.
- We specialize relative-entropy tilting to phase utilities measured by a synchronized learned reward and anchor it to on-policy occupancy.
- We derive an occupancy-corrected batch reward whose immediate aggregation matches the phase objective, then integrate it with PPO without manually designed tracking or phase rewards.

II. Related Work

A. Reference Motion Control

DeepMimic showed that deep reinforcement learning can reproduce diverse motion-capture clips in simulation [1]. Its reward is a weighted combination of pose, velocity, end-effector, root, and center-of-mass tracking. Subsequent systems scale tracking to motion collections, improve recovery, and support more general control through new architectures and training procedures [2], [3]. These advances make high-quality character control possible, but manually composed tracking objectives remain a significant part of the specification. ScaDiver scales control through a mixture-of-experts architecture [4], UniCon separates motion scheduling from low-level execution [5], and PULSE distills a reusable latent motion representation [6]. These works address controller capacity, representation, and recovery. PLOT is complementary: it asks whether a brief low-quality interval can be offset by higher reward elsewhere.

B. Adversarial Imitation and Learned Rewards

GAIL casts imitation as adversarial occupancy matching [7]. AMP applies adversarial learning to physics-based characters by discriminating short state transitions and using the learned signal as a style prior [8]. These methods

DRAFT ASSET—replace each panel with a synchronized multi-frame sequence

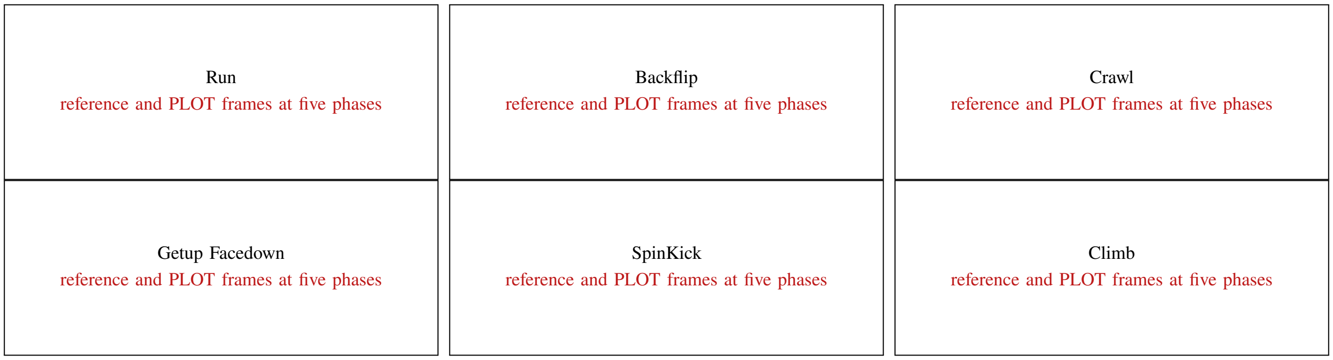

Fig. 1. Reference motions and corresponding policy rollouts, synchronized at common phases.

learn distributional notions of natural motion, but frame-synchronized tracking requires a reward that distinguishes the realized character from a particular reference at the same time.

Other work changes how an adversarial reward is inferred or regularized. AIRL targets rewards that remain meaningful under dynamics changes [9]; a variational discriminator bottleneck constrains information flow to preserve useful gradients [10]; and the divergence-minimization view unifies several adversarial imitation objectives [11]. PLOT instead holds the learned discriminator and its reward mapping fixed.

Frame-synchronized differential discriminators instead compare the simulated character with its reference at the same instant, mapping perfect tracking at every frame to a common zero residual [12]. This construction provides a local score tied to a prescribed clip and learns correlations among pose and velocity discrepancies without separate coefficients. PPO still receives one scalar per transition, however. The discriminator determines which synchronized transition resembles ideal tracking; the return determines how these local judgments trade across the clip. PLOT changes only the latter.

C. Multi-Objective and Robust Policy Optimization

Multi-objective reinforcement learning studies policies under several possibly competing objectives and the preferences used to combine them [13]. Linear scalarization can conceal weak objectives; distributionally robust optimization instead reweights high-loss observations within an ambiguity set [14]. Tilted empirical risk minimization provides an established log-sum-exp family between average and extremal loss [15]. In imitation learning, PG-BROIL applies a soft-robust policy objective to uncertainty over learned reward hypotheses [16].

PLOT does not claim exponential tilting as a new optimization primitive. Its contribution is to place the ambiguity distribution over reference phases of one synchronized learned reward, anchor it to on-policy phase occupancy, and realize the resulting density ratio before PPO advantage estimation. Unlike a sampling curriculum, this construction leaves data collection unchanged; and unlike reward-robust imitation, it introduces neither multiple reward hypotheses nor manually weighted physical objectives.

III. Background

Let s_t be the simulated character state, $\hat{s}_t$ the synchronized reference state, and o_t the observation of a policy $\pi_\theta(a_t \mid o_t)$. Applying action a_t advances the simulator to s_{t+1}. The tracking problem seeks a policy that reproduces the entire reference sequence while satisfying the simulated dynamics.

Classical tracking rewards first define separate errors. Denote pose, velocity, root, end-effector, and contact errors by E^p_t, E^v_t, E^r_t, E^e_t, and E^c_t. A representative reward is

$$r^{\text{hand}}_t = \sum_{i\in\{p,v,r,e,c\}} w_i \exp\bigl(-\alpha_i E^i_t\bigr), \tag{1}$$

where w_i sets relative importance and α_i sets sensitivity. Joint- or motion-specific weights, contact schedules, survival bonuses, and termination rules can add further design decisions. Because the errors have different units and distributions, Eq. (1) is not merely a numerical convenience; it defines which imperfect behaviors are preferred.

Replacing Eq. (1) with a learned reward removes *manual feature-wise scalarization*. Ordinary reinforcement learning nevertheless maximizes the discounted sum of learned rewards and therefore still performs *phase-wise scalarization*: each region of the reference contributes in proportion to its rollout frequency. Low reward during a brief but necessary transition can consequently be offset by high reward elsewhere. This temporal compensation is the problem addressed below.

The implicit temporal weights can be made explicit at the rollout level. Let r^ℓ_t be any learned tracking reward in a collected batch $\mathcal{B}$ of N transitions. Suppose the reference clock is divided into K intervals and $c_t \in \{1,\ldots,K\}$ identifies the interval reached by transition t. If N_j is the number of samples in interval j, define the observed support $\mathcal{I}_\mathcal{B} = \{j : N_j > 0\}$. For $j \in \mathcal{I}_\mathcal{B}$, empirical occupancy is $\rho_j = N_j/N$ and mean learned reward is $J^\ell_j = N_j^{-1}\sum_{t\in\mathcal{B}:c_t=j} r^\ell_t$. The batch mean therefore decomposes exactly as

$$\frac{1}{N}\sum_{t\in\mathcal{B}} r^\ell_t = \sum_{j\in\mathcal{I}_\mathcal{B}} \rho_j J^\ell_j. \tag{2}$$

Thus, replacing handcrafted feature terms by a learned scalar does not remove scalarization; it moves the remaining choice to time. The standard return uses the phase occupancy induced jointly by initialization, termination, horizon, and the current policy. A long easy interval receives more total influence than a short difficult interval even when the latter is essential to completing the reference. Discounting can further favor intervals encountered earlier within an episode. This behavior is appropriate when rewards are freely substitutable, but ordered motion imitation has a conjunctive aspect: takeoff, rotation, and landing cannot generally replace one another.

The occupancy ρ_j is not a property of the reference clip alone. It also depends on how episodes are initialized, how long they survive, whether the clip wraps, and which states the evolving controller reaches. Early failure can remove later phases from a batch, while reference-state initialization can make the clock nearly uniform even when the physical behavior is poor. Moreover, temporal compensation persists under uniform occupancy: an arithmetic mean still permits a large utility at one interval to offset a small utility at another. Correcting occupancy and emphasizing weak utility are consequently different operations. The former asks how much total mass each observed phase receives; the latter asks which distribution over those phases represents the desired temporal objective.

We group the learned scalar reward rather than the underlying physical errors. This distinction preserves the reason for learning the reward in the first place. A phase objective defined separately over position, orientation, and velocity would reintroduce a choice of units and tradeoffs at every phase. In contrast, conditional means of one shared learned reward remain comparable within a rollout and expose only the temporal aggregation decision. They are not semantic success labels: a low J_j^{ℓ} simply indicates that the same learned tracking criterion assigns lower utility to transitions synchronized with interval j.

Standard PPO subsequently applies discounting and advantage estimation to this same base learned-reward sequence. These operations propagate credit across time, but they do not introduce a criterion requiring weak reference phases to improve. PLOT changes the pre-GAE aggregation in Eq. (2) and then delegates long-horizon credit assignment to the unchanged RL update.

Feature-wise and phase-wise aggregation are therefore orthogonal. The learned reward determines what constitutes a good synchronized transition; temporal aggregation determines how improvements at different reference times compete for a finite policy update. PLOT retains the former and replaces the nominal occupancy weights in Eq. (2) by a smooth, utility-aware distribution. Its implementation uses empirical batch analogues of these quantities before constructing discounted returns, as described next.

IV. PLOT: Phase-Level Objective Tilting

A. Learned Adversarial Reward

We first learn a single instantaneous tracking reward instead of manually combining pose, velocity, root, and contact terms. Let $g(s) \in \mathbb{R}^d$ denote a common character representation and let $\mathcal{N}$ be a scale-only normalizer. The discriminator observes the normalized synchronized residual $\bar{\mathbf{e}}_t = \mathcal{N}(g(\hat{s}_t) - g(s_t))$. Perfect tracking produces the same zero residual at every reference frame, whereas normalized residuals from the current policy and replay buffer form a negative distribution q_-. A discriminator produces a scalar logit $f_\psi : \mathbb{R}^d \to \mathbb{R}$ and the ideal-class probability $D_\psi(\bar{\mathbf{e}}) = \sigma(f_\psi(\bar{\mathbf{e}}))$, where σ is the logistic sigmoid.

For the humanoid, g is a one-frame, 172-dimensional description containing root translation and orientation, root linear and angular velocity, joint rotations, joint velocities, and root-relative body positions. Quaternion orientations are mapped to a continuous tangent–normal representation before subtraction, avoiding the discontinuity of wrapped Euler angles. Reference and simulated states pass through the same feature map and coordinate convention; consequently, each component of $g(\hat{s}_t) - g(s_t)$ compares like with like. Global root quantities preserve the displacement and rotation needed for locomotion and acrobatics, while root-relative body positions describe the articulated configuration without duplicating global translation across every body. The representation includes both configuration and velocity because two characters at the same pose can require different control when their momenta differ.

The normalizer $\mathcal{N}$ is scale-only: it divides each residual coordinate by its running mean absolute magnitude and leaves the origin fixed. This makes the discriminator optimization less sensitive to physical units without introducing an offset or a hand-selected coefficient for a feature group. In particular, normalization does not prescribe the tradeoff among a root-position error, a joint-rotation error, and a velocity error. That tradeoff is represented by the discriminator learned from the policy's own residual distribution. Current-rollout and replay residuals contribute equally to q_-, so the negative class reflects both the present controller and recently visited errors.

Mapping every ideal synchronized pair to the common origin removes the need to learn a separate positive manifold for each reference frame. At the same time, the residual is not reduced to a scalar norm: its signed coordinates preserve which bodies lead or lag the reference and which velocities disagree. The discriminator can therefore represent correlations such as an orientation error that is benign at one angular velocity but incompatible at another, without an explicit manually weighted decomposition. This learned geometry is instantaneous and frame synchronized. It establishes the base reward used by PLOT but, by itself, does not specify how rewards from separate phases should compete in a finite policy update.

Assigning label one to the zero residual and label zero to policy residuals, then taking the expected Bernoulli log-likelihood under equally likely classes, gives the balanced classification objective

$$\max_{\psi} \ \mathcal{J}_D^{\text{cls}} = \frac{1}{2} \log D_\psi(\mathbf{0}) + \frac{1}{2} \mathbb{E}_{\bar{\mathbf{e}} \sim q_-} \log\left[1 - D_\psi(\bar{\mathbf{e}})\right]. \quad (3)$$

The two likelihood terms learn how heterogeneous residual components should be aggregated when separating ideal and policy samples. In training, we minimize the negative of Eq. (3) together with the inherited zero-centered logit-gradient penalty $\lambda_{\text{gp}}\mathbb{E}_{q_-}[\|\nabla_{\bar{\mathbf{e}}} f_\psi(\bar{\mathbf{e}})\|_2^2]$ and the output head's inherited ℓ_2 weight regularizer. These terms discourage an excessively sharp boundary and uncontrolled logit scale; neither defines a tracking preference.

Equation (3) follows directly from balanced Bernoulli maximum likelihood. The positive distribution is analytically available as the point mass at $\mathbf{0}$; it is not a collection of privileged reference states. The negative distribution is induced by the policy and therefore changes as tracking improves. At its optimum, the classifier estimates how separable the current residual distribution is from perfect tracking. This construction is important for removing handcrafted rewards: the labels specify only whether a residual is ideal, not how the coordinates should be weighted or what numerical distance should count as success.

The policy is rewarded for moving its residual toward the ideal class. Let $s_r > 0$ be a global reward scale. After the simulator produces s_{t+1}, we use the negative log-likelihood of the policy class,

$$r_t^{\text{adv}} = -s_r \log\left[1 - D_\psi(\bar{\mathbf{e}}_{t+1})\right]. \tag{4}$$

Equation (4) contains no separate coefficient for pose, velocity, contact, or completion. Neither the physical error metrics nor the evaluation completion tests are available during training.

The reward and discriminator are two sides of the same classification game. The negative likelihood in Eq. (3) pushes $D_\psi(\bar{\mathbf{e}})$ toward zero on policy residuals, whereas maximizing Eq. (4) pushes the policy toward residuals assigned higher ideal-class probability. Using the discriminator logit, the reward is equivalently $s_r\,\text{softplus}(f_\psi)$; this identity is an implementation of the negative-class log-likelihood, not an additional tracking term. PLOT preserves both this objective and this reward mapping. Its only algorithmic change begins after the scalar rewards have been computed.

The actor and discriminator have distinct interfaces. The discriminator sees only the synchronized residual described above. The actor uses exactly the same interface as the matched learned-reward baseline: a physical self-observation concatenated with target observations at the next three control offsets. Future root displacement is encoded relative to the current simulated root, together with future root orientation, joint rotations, and key body positions. This short causal lookahead disambiguates motions that share a present configuration but evolve differently and may itself reveal temporal context. PLOT does not append an explicit phase index or phase weight to either network; it changes the objective rather than the observation. The actor outputs joint-position targets for the 28 actuated degrees of freedom; proportional–derivative controllers convert these targets into physical actuation. PLOT leaves this action space, controller, actor, critic, and discriminator architecture unchanged.

B. Phase-Level Objective Tilting

Let $\tau_t \in [0,1)$ denote normalized reference phase. We divide the reference clock into K control-rate intervals and assign each post-step reward to $c_t = 1 + \lfloor K\tau_{t+1}\rfloor$, clamped to $\{1,\ldots,K\}$. At the start of a policy update, a frozen behavior rollout $\mathcal{B}$ supplies the observed support $\mathcal{I}_\mathcal{B}$ defined in Section III. Its empirical occupancy is $\rho_j = N_j/|\mathcal{B}|$, and its utility J_j is the mean of r_t^{adv}/s_r over those transitions. Both are fixed batch statistics during the update, and every J_j is measured by the same discriminator.

Let $\mathbf{p}$ be a candidate distribution on the simplex over $\mathcal{I}_\mathcal{B}$ and let $\beta > 0$ set the strength of the tilt. Writing KL for Kullback–Leibler divergence, PLOT defines the batch phase aggregation

$$\Phi_\beta(\mathbf{J}) = \min_{\mathbf{p}\in\Delta(\mathcal{I}_\mathcal{B})} \left\{ \sum_{j\in\mathcal{I}_\mathcal{B}} p_j J_j + \frac{1}{\beta}\text{KL}(\mathbf{p}\|\rho) \right\}. \tag{5}$$

The inner distribution evaluates the policy more strongly where learned utility is low. Relative entropy anchors this tilt to the rollout distribution; without it, a noisy estimate of the single worst interval would determine the entire objective.

The KL penalty prices departure from the frozen empirical occupancy; it does not assert that the rollout itself came from an adverse phase distribution. The discriminator still defines tracking quality, while $\mathbf{p}$ determines where that quality enters the batch surrogate. Anchoring to ρ avoids amplifying a sparsely observed phase solely because it is rare. If all utilities are equal, $\mathbf{p} = \rho$ and PLOT recovers the standard learned-reward update.

The inner problem is convex. Stationarity of its Lagrangian followed by simplex normalization gives

$$\begin{aligned} p_j^* &= \frac{\rho_j \exp(-\beta J_j)}{\sum_{k\in\mathcal{I}_\mathcal{B}} \rho_k \exp(-\beta J_k)}, \\ \Phi_\beta(\mathbf{J}) &= -\frac{1}{\beta}\log\left(\sum_{j\in\mathcal{I}_\mathcal{B}} \rho_j \exp(-\beta J_j)\right). \end{aligned} \tag{6}$$

This Gibbs form is the exponential tilt induced by the variational objective [17]. As $\beta \to 0$, $\mathbf{p}^* \to \rho$ and the ordinary temporal average is recovered. Equivalently, the phase-space density ratio tends to one, so the actual update recovers standard learned-reward PPO. Larger β places more relative mass on lower-utility intervals. The intervals follow the control-rate reference clock and carry no semantic labels such as takeoff, landing, or contact.

Introducing a multiplier for $\sum_j p_j = 1$ gives $\log(p_j/\rho_j) = -\beta J_j$ up to a shared constant. Thus two phases with equal utility retain their nominal mass ratio, and the log-sum-exp denominator avoids the active-phase switching of hard max–min training.

The effect on individual transitions is especially transparent when comparing two observed intervals i and j: their density-ratio quotient is $(p_i^*/\rho_i)/(p_j^*/\rho_j) = \exp[-\beta(J_i - J_j)]$. If interval i has utility lower by $\Delta > 0$, each of its samples has $\exp(\beta\Delta)$ times the temporal weight of a sample from interval j. The occupancies cancel in this per-sample comparison.

Thus an interval is not amplified merely because it is rare, but a brief interval that is both observed and poorly tracked can exert greater influence than its duration would provide under an ordinary average. As its utility catches up, the ratio returns continuously toward one. This negative feedback is the mechanism by which PLOT opposes temporal compensation: repeatedly collecting high reward on easy regions does not erase the relative deficit at a decisive region before the policy update is formed.

This behavior differs from three nearby objectives. Standard learned-reward PPO sets $w_t = 1$ and therefore retains whatever phase mass the rollout supplies; it has no weak-phase response. Uniform-phase averaging chooses $p_j = 1/K$, which uses inverse occupancy regardless of tracking quality. It can over-amplify a rare but already well-tracked interval and can produce large weights from sampling noise alone. A hard minimum places all mass on the current lowest estimate. That objective prevents compensation most aggressively, but its active interval can switch after an arbitrarily small estimation change, so successive PPO batches may optimize different bottlenecks. PLOT preserves the support-aware nominal measure of the first method, the phase balancing intent of the second, and the weak-phase sensitivity of the third, while replacing their abrupt or occupancy-only choices with one smooth density ratio.

The soft minimum obeys $\min_{j\in\mathcal{I}_{\mathcal{B}}} J_j \le \Phi_\beta(\mathbf{J}) \le \sum_{j\in\mathcal{I}_{\mathcal{B}}} \rho_j J_j$. The left limit is approached as β grows, whereas the right limit is approached as β vanishes. Finite β therefore limits how much one noisy interval can dominate without restoring unrestricted compensation. The objective is also invariant, up to an additive shift, to a common translation of all J_j values; the weights depend on gaps rather than on the absolute discriminator reward offset. Together with mean-one density ratios, these properties preserve the scale of the PPO signal more predictably than either hard switching or unconstrained inverse-frequency weighting.

Expanding around zero gives $p_j^*/\rho_j = 1 - \beta(J_j - \sum_k \rho_k J_k) + O(\beta^2)$, which quantifies the smooth departure from ordinary return. Because $\mathbf{p}$ remains absolutely continuous with respect to ρ, however, the method cannot place mass on a phase absent from the rollout. Broad reference-state initialization is therefore complementary rather than replaced by objective tilting.

Scaling behaves differently from translation: multiplying all utilities by a positive constant b is equivalent to replacing β by $b\beta$. We therefore compute J_j after removing the configured global reward scale s_r, so changing that scalar does not silently change the strength of temporal robustness. This normalization does not claim that independently trained discriminators are calibrated across motions; it only separates the PLOT temperature from a known global multiplier.

Finally, the control-rate partition supplies indices, not semantic supervision. No labels identify a phase as difficult, and neighboring intervals are not assigned handcrafted similarity. Their utilities are inferred from the same reward and the same on-policy transitions as all other intervals. This is sufficient for ordered single-reference motions because synchronization already provides a clock. The phase index is retained as batch metadata and is never exposed as an actor command or discriminator feature.

C. Policy Optimization

The target phase distribution $\mathbf{p}^*$ generally differs from the frozen empirical occupancy. A transition assigned to interval c_t receives

$$w_t = \frac{p^*_{c_t}}{\rho_{c_t}}, \qquad \widetilde{r}_t = w_t r_t^{\text{adv}}. \tag{7}$$

The ratio is a phase-space density ratio—a change of measure over reference time—rather than a new phase reward or an off-policy importance ratio. Substituting Eq. (6) yields $w_t = \exp(-\beta J_{c_t})/\sum_{k\in\mathcal{I}_{\mathcal{B}}} \rho_k \exp(-\beta J_k)$: low occupancy alone does not increase a transition's influence; its learned utility must also be low. The rollout mean of w_t is one, so the construction redistributes the immediate reward without a trivial global rescaling; discounting and GAE can still change the distribution of policy advantages.

For the frozen batch, this change of measure exactly matches the inner aggregation at the level of immediate rewards:

$$\sum_{j\in\mathcal{I}_{\mathcal{B}}} \rho_j \left(\frac{p_j^*}{\rho_j}\right) J_j = \sum_{j\in\mathcal{I}_{\mathcal{B}}} p_j^* J_j. \tag{8}$$

Without division by ρ_j, sampling frequency would enter twice: once through the number of transitions in phase j and again through p_j^*. Conversely, using an inverse-frequency weight alone would implement uniform-phase averaging rather than the utility-dependent objective. Equation (8) is the bridge from Eq. (5) to the batch reward sequence. It does not imply that discounted PPO exactly maximizes Φ_β: PPO subsequently optimizes its usual clipped, discounted-return surrogate using the frozen reweighted rewards.

In the algorithm, we denote the frozen batch statistics by $\widehat{J}_j$, $\widehat{\rho}_j$, and $\widehat{p}_j^* \propto \widehat{\rho}_j \exp(-\beta \widehat{J}_j)$. The resulting $\widehat{w}_t = \widehat{p}^*_{c_t}/\widehat{\rho}_{c_t}$ and the discriminator are frozen throughout the policy epochs. We compute generalized advantage estimation (GAE) from $\widetilde{r}_t = \widehat{w}_t r_t^{\text{adv}}$ and apply the standard clipped PPO update [18], [19]. Reweighting precedes return construction, so a low-utility landing can alter the advantages of earlier takeoff and rotation actions through ordinary temporal credit assignment. The discriminator is updated after the policy epochs, and the next rollout produces a new tilt. This alternating procedure is a batch surrogate for Eq. (5), not an exact end-to-end minimax gradient. Intervals absent from a finite batch are excluded from that iteration's normalization.

We apply the phase-space density ratio before GAE rather than to isolated policy-gradient terms. Many decisive events are determined by earlier actions: angular momentum for a flip is accumulated before inversion, and a successful landing depends on the takeoff state. For a rollout horizon H, PLOT first constructs $\widetilde{r}_{0:H-1}$ and then runs the usual return recursion. A weak future phase can thereby alter the advantages assigned to its causal predecessors. This changes the reward measure but retains the established discounting,

Algorithm 1 PLOT training

Require: reference motion and tilt parameter β

Initialize policy π_θ, value function V_ω, discriminator D_ψ, and replay buffer
while training budget remains **do**
  Collect rollout $\mathcal{B}$; store residuals and post-step phases
  Evaluate r_t^{adv} for every transition in $\mathcal{B}$
  Estimate observed phase utilities $\widehat{J}_j$ and occupancies $\widehat{\rho}_j$
  Compute $\widehat{p}_j^* \propto \widehat{\rho}_j \exp(-\beta \widehat{J}_j)$
  for transition $t \in \mathcal{B}$ **do**
    $\widetilde{r}_t \leftarrow (\widehat{p}_{c_t}^* / \widehat{\rho}_{c_t}) r_t^{\mathrm{adv}}$
  end for
  Compute GAE from $\widetilde{r}_t$; update π_θ, V_ω by PPO
  Update D_ψ with ideal, rollout, and replay residuals
end while

value baseline, and PPO clipping that perform long-horizon credit assignment.

Because utilities, occupancy, and the discriminator all change with the policy, this frozen-batch update is not an exact global minimax gradient and does not differentiate through the phase statistics or discriminator. Only the inner tilt is solved exactly for each sampled batch; physical policy improvement retains the approximation and exploration limits of PPO.

The surrogate has a deliberate scope. It assumes an ordered, synchronized reference for which a control-rate clock is meaningful; an unordered motion prior has no analogous phase axis. For a motion collection, utilities and occupancies must be formed within each selected reference rather than pooled across unrelated clips. Because the tilted distribution is supported only on observed phases, PLOT does not replace exploration or reference-state initialization. It also cannot repair a base discriminator that assigns high reward to an incorrect behavior at every phase. Finally, tilting one learned scalar does not guarantee that every physical error decreases: completion, position, rotation, and velocity must remain separate evaluation outcomes.

V. Experimental Setup

A. Motions, Baselines, and Protocol

We evaluate six single-clip skills: Run, Backflip, Crawl, Getup Facedown, SpinKick, and Climb (Fig. 1). They span sustained locomotion, aerial rotation, prolonged ground contact, recovery, rapid striking, and ordered interaction with a rigid box. The suite contains both errors distributed throughout a clip and short transitions whose omission changes the identity of the motion.

We compare four methods. **DeepMimic** uses a manually weighted tracking reward [1]. **AMP** learns an adversarial motion prior from state transitions [8]. **ADD** uses a learned frame-synchronized differential reward with its published absolute reference interface [12]. **PLOT** uses exactly the same 464-dimensional actor input, discriminator input, reward, and network architecture as ADD and changes only the temporal aggregation in Eq. (5). An additional uniform-phase ablation ($p_j = 1/K$) separates utility-dependent tilting from generic occupancy correction. These matched interfaces permit direct attribution of PLOT–ADD differences to the temporal objective.

Each formal policy is trained for 2000 iterations with 8192 parallel environments. Control runs at 30 Hz and simulation at 120 Hz. Network, action, PPO, optimizer, and discriminator settings are shared across the learned-reward methods. We fix $\beta = 1.90$ for every evaluated motion after selecting it on separate Roll development runs; Roll is excluded from the six-motion result table. The current protocol uses one training seed, so the results support controlled per-motion comparisons rather than estimates of seed-level variance.

Random resets sample the reference clock during training. Pose-error termination is disabled, so pose deviation alone does not end an episode. The shared physical termination based on disallowed contacts remains active, but the allowed-contact specification generates no reward, and the coefficient of any task-specific reward is zero. Climb includes the same interactable box for every method.

B. Evaluation Metrics

We evaluate deterministic phase-zero rollouts without automatic reset. Wrapped motions run to a fixed horizon and non-wrapped motions to their natural clip endpoint. The vectorized replicas share an initial condition and are therefore used for evaluation throughput, not as independent statistical trials. We report body-position error E_{pos} in meters, body-rotation error E_{rot} in radians, and joint-velocity error E_{vel} in radians per second.

Because time-averaged errors do not establish that an ordered motion is complete, we also report a binary completion outcome. The evaluator combines survival with motion-appropriate sequence checks, including displacement, rotation winding, recovery posture, or climbing progress. These criteria are fixed before evaluation and never enter training or checkpoint selection. To test the proposed mechanism, we additionally examine the learned utility over reference phase and the corresponding PLOT density ratio.

VI. Results

A. Tracking Fidelity and Motion Completion

Table I reports the matched final-checkpoint evaluations. DeepMimic and AMP each complete four of six motions, whereas ADD completes five and fails Climb. Across all six motions, their respective mean $(E_{\mathrm{pos}}, E_{\mathrm{rot}}, E_{\mathrm{vel}})$ values are $(0.109, 0.351, 0.867)$, $(0.202, 0.569, 1.581)$, and $(0.062, 0.165, 0.706)$. These descriptive averages span heterogeneous skills; the per-motion values and completion outcomes remain the primary comparison.

On Run, PLOT and ADD both satisfy the completion rule. PLOT obtains $(.031, .088, .943)$, compared with ADD's $(.032, .087, .946)$. Using the unrounded evaluation artifacts, the PLOT position and joint-velocity errors are lower by approximately 4.5% and 0.3%, respectively, while rotation error is about 1.0% higher. We retain all three components rather than collapsing this mixed outcome into a favorable scalar. The other PLOT entries are explicitly marked as unmeasured; this one cyclic locomotion result does not establish an advantage over the motion suite.

Completion and tracking error answer different questions. A method cannot offset omission of an ordered event with low average error on the phases it does execute, while a completed motion is not automatically accurate. We therefore report both at fixed checkpoints under one protocol; PLOT is successful only when reduced temporal compensation is reflected in completion or physical tracking, not merely in its training surrogate.

B. Temporal Aggregation Analysis

The core ablation holds the learned reward, policy observation, network, and PPO settings fixed and changes only temporal aggregation. Standard learned reward uses $w_t = 1$. Uniform-phase correction assigns equal total mass to each observed interval, separating occupancy correction from utility-dependent tilting. PLOT additionally adapts that mass through Eq. (6). This comparison isolates the proposed temporal mechanism from the end-to-end differences among the systems in Table I.

The mechanism is directly testable along the reference clock. If utilities are nearly equal, Eq. (6) remains close to the rollout occupancy and w_t remains near one. A localized utility deficit should instead produce a larger density ratio in the corresponding interval. Improvement of that interval should be accompanied by better physical tracking or completion, rather than merely by a change in discriminator scale.

Run provides a useful negative-control case for this mechanism. At its final training iteration, the normalized entropy of the tilted phase distribution is 0.998 and the largest transition weight is 1.317. Thus PLOT remains close to nominal occupancy once this cyclic motion has similar utility across its clock; it does not manufacture a persistent hard bottleneck. The modest tilt is consistent with the matched completion and mixed, percent-level error changes in Table I. A decisive acrobatic or interaction motion must show a more localized utility deficit and a corresponding physical improvement before the proposed anti-compensation mechanism is supported.

The matched ablation makes the claim falsifiable. Standard-to-uniform gains would support occupancy correction, while an additional PLOT gain would isolate utility-dependent tilting. A higher minimum learned utility without improved completion or physical error counts against the mechanism; improved completion with worse mean error is reported as a robustness–fidelity tradeoff, not uniform superiority.

[TBD: report the measured Standard/Uniform/PLOT ablation and any completion–fidelity tradeoff].

VII. Conclusion

We introduced Plot, a phase-level objective for motion control with learned adversarial rewards. Whereas the discriminator aggregates heterogeneous tracking discrepancies within a frame, Plot addresses compensation across reference phases by exponentially tilting rollout occupancy toward lower learned utility. The resulting density ratio reweights the same base learned reward before advantage estimation and requires neither a manually designed phase reward nor an additional learned model.

[TBD: insert one suite-level completion result, one tracking result, and the matched temporal-objective ablation]. The current single-seed protocol does not support claims about random-seed variance, and phase tilting cannot correct distinctions that the learned reward itself fails to represent. Within these limits, PLOT isolates temporal aggregation as a distinct design problem after handcrafted feature rewards have been removed.

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TABLE I

FINAL-CHECKPOINT PERFORMANCE UNDER DETERMINISTIC EVALUATION. COMPLETE IS A BINARY FULL-MOTION OUTCOME; ERRORS ARE TRANSITION MEANS. RED ENTRIES HAVE NOT YET BEEN MEASURED.

Motion	Method	Complete	E_{pos}	E_{rot}	E_{vel}	Motion	Method	Complete	E_{pos}	E_{rot}	E_{vel}
Run	DeepMimic	1	.028	.106	1.096	Crawl	DeepMimic	1	.008	.045	.252
	AMP	1	.068	.171	1.484		AMP	1	.090	.281	.627
	ADD	1	.032	.087	.946		ADD	1	.011	.053	.161
	PLOT	1	.031	.088	.943		PLOT	**TBD**	**TBD**	**TBD**	**TBD**
Backflip	DeepMimic	0	.177	.605	1.303	Getup	DeepMimic	1	.026	.104	.408
	AMP	0	.435	1.080	2.779		AMP	1	.173	.476	1.503
	ADD	1	.061	.187	1.091		ADD	1	.030	.123	.324
	PLOT	**TBD**	**TBD**	**TBD**	**TBD**		PLOT	**TBD**	**TBD**	**TBD**	**TBD**
SpinKick	DeepMimic	1	.027	.117	1.245	Climb	DeepMimic	0	.385	1.131	.896
	AMP	1	.066	.169	1.716		AMP	0	.382	1.238	1.375
	ADD	1	.031	.112	.965		ADD	0	.205	.430	.749
	PLOT	**TBD**	**TBD**	**TBD**	**TBD**		PLOT	**TBD**	**TBD**	**TBD**	**TBD**

TBD measured completion curve

TBD measured position error

TBD measured rotation error

TBD measured joint-velocity error

Fig. 2. Checkpoint evaluation over training. The four panels report full-motion completion and the three physical tracking errors under the fixed evaluation protocol. Curves will show measured checkpoints only.

TBD measured phase utility by reference phase

TBD measured PLOT density ratio by phase

Fig. 3. Phase-level mechanism analysis for a decisive motion. Left: learned utility along the reference clock. Right: the density ratio applied to the same rollout.

TABLE II

MATCHED TEMPORAL-OBJECTIVE ABLATION. "WORST" DENOTES THE MINIMUM NORMALIZED PHASE UTILITY. HIGHLIGHTED ENTRIES AWAIT FORMAL EVALUATION.

Aggregation	Complete	E_{pos}	Worst J_j	max w_t
Standard ($w = 1$)	**TBD**	**TBD**	**TBD**	1.0
Uniform phase	**TBD**	**TBD**	**TBD**	**TBD**
PLOT	**TBD**	**TBD**	**TBD**	**TBD**